

PROJECT AND TEAM INFORMATION

Project Title

Orbital Collision Avoidance System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

LEO's (Lower Earth Orbit) already at breaking point We have thousands of collision alerts fired to satellite operators every single day – and not many are of real concern. Traditionally systems simply calculate the minimum distance to another object, but simple distance does not tell the whole story. You need to integrate multiple variables and uncertainty estimates to determine if a danger to collision is real.

And when you do: does your satellite have the fuel and manoeuvre capability to escape the predicted trajectory of the object.

You only need to look at the Iridium 33 and Cosmos 2251 collision in 2009 – and all of the fragments still littering LEO. And with over 4000 new payloads launched on the orbit on a given year – and many with the same orbital path - ESA (European Space Agency) forecasts collisions will soon happen four times more often than are sustainable! Current ground-based solutions have been stretched beyond limit and cannot cope with volume of traffic to keep our space environment safe – we must have an on board autonomous system to solve these calculations.

State of the Art / Current solution

Right now, our orbital outposts only really trust sentinels on the surface of the planet to give them the heads-up if a particularly large debris is about to smush into them. Tens of thousands of fragments are monitored by the USSF, who then beam the raw data back to NASA, where humans analyze each one carefully. Even when we introduce something like the ESAs CREAM, which uses machine learning principles to help weed out the false positives, decisions with actual life or death stakes are still being calculated at Earth-based servers.

Researchers at the university level are also optimizing the maneuvers themselves, but usually are concentrating on one specific class of rocket engine. There's some discussion of purely optical sensors similar to designs patented for drones, but real-world autonomy is more conceptual.

Out in space, space flight software is typically quite unyielding: the satellite will simply trigger thrusters if some orbital bit has exceeded some certain distance or has some 1- in-10,000 chance of collision, and will continue burning under this one criterion. So we're holding ourselves captive back on Earth; communications delay, simple-logic decision-making and a lack of hardware agility mean our current ground-to-space pipelines create an immense bottleneck.

Project Goals and Milestones

The main goal of this project is to illustrate and assess conjunction risk and decide which treat should be executed first

Initial milestones:

- 1. Study existing collision avoidance systems.*
- 2. Analyse a LEO conjunction dataset.*
- 3. Implement circular and priority queues.*
- 4. Develop a basic Logistic Regression model.*
- 5. Develop the C++ thruster classes.*
- 6. Add fuel, burn-time and TCA checks.*
- 7. Add fallback, rollback and logging.*
- 8. Integrate and test the complete system.*

Project Approach

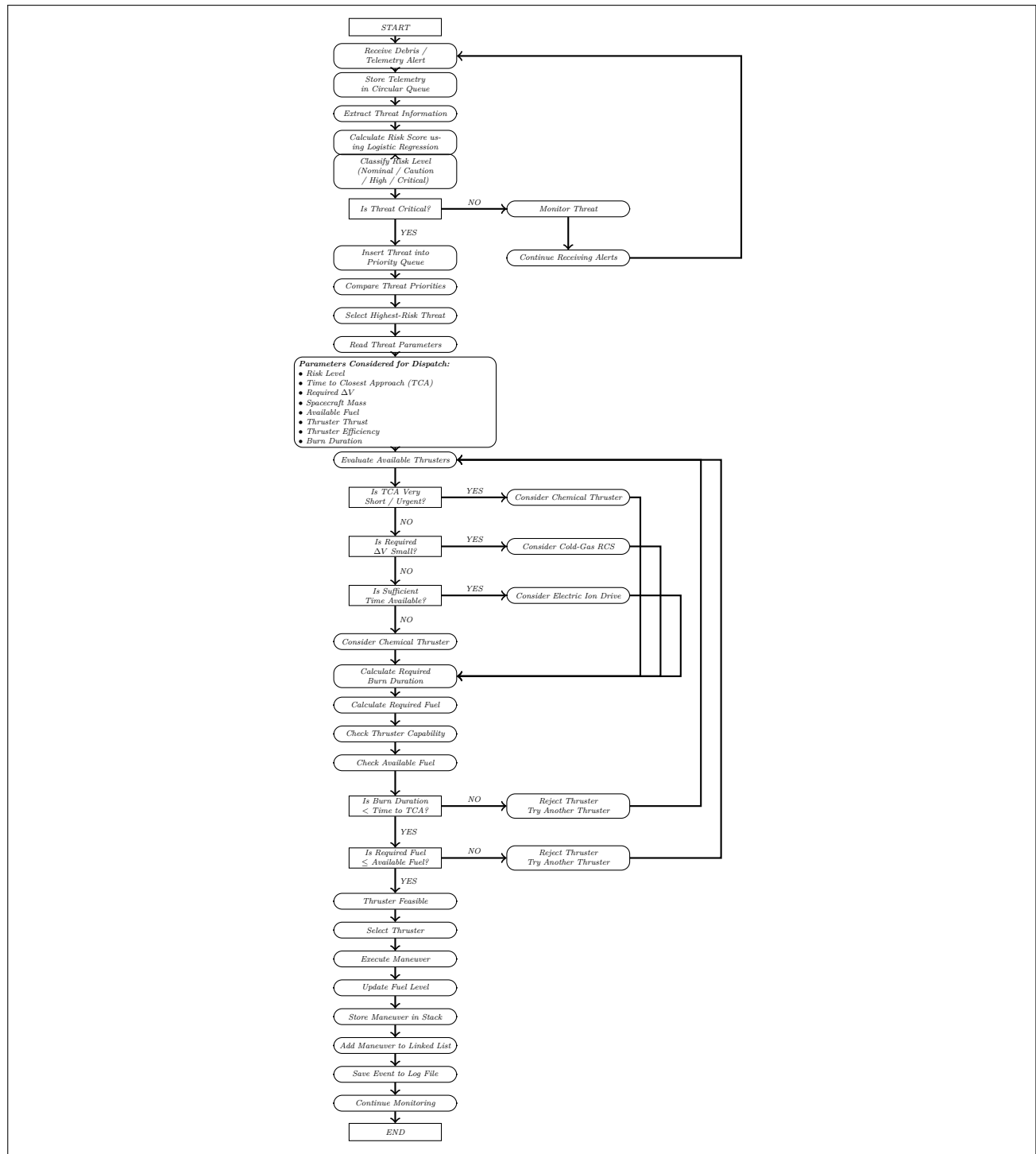
We will create an integrated system where we integrate ML, Data Structures, and Object-Oriented Programming to determine space object collision risks and the appropriate actions to perform collision avoidance. Initially, we will investigate the existing techniques in use to track orbiting bodies and detect a conjunction(space object threat) and predict suitable collision avoidance measures. Based on the researches and shortcomings, we will collect conjunction data from the dataset available online, analyze the data, and extract essential parameters required to conclude decisions like minimum separation distance, relative velocity, time of closest approach (TCA), and the probability of collision.

A Circular Queue, a Queue, and a Priority Queue data structures will be integrated in our system to maintain the influx of telemetry data and threat alerts. A Logistic Regression model will then be trained to generate a risk score (based on multiple features and not just distance alone) from each threat alert. The risk scores generated by the model will be utilized to prioritize threats so that we can process the most critical threats first.

The propulsion mechanism will be designed using OOPS with a generic ThrusterEngine class that contains specific thruster types like Chemical, Electric Ion, and Cold-Gas Thruster. By comparing criteria such as the required change in velocity (ΔV), burn time, fuel availability, and time to closest approach, a suitable thruster will be selected. If the chosen thruster is not optimal, another eligible thruster will be considered before proceeding with the maneuver.

Lastly, successfully completed collision-avoidance maneuver records will be managed using stack and linked list data structures, and relevant log data will be written into a log file. The system developed will be tested using simulated threat and system failure conditions to check for its accuracy in predicting safe maneuvers.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

OCAS has five main technical components:

- *A flight software prototype written in C and C++ that uses Circular and FIFO queues for memory management, and a C++ OO dispatcher, for telemetry that is buffered without allocation.*
- *A machine learning model that uses Logistic Regression and is trained to classify risk of conjunctions in LEO. The model is able to identify encounter geometries and assign them a risk probability.*
- *A threat arbitration engine that uses a C based Max-Priority Queue (Binary Heap). This engine is used to prioritize threats.*
- *A hardware agnostic module that simulates thruster functional limitations, and using Tsiolkovsky's Rocket Equation determines the feasibility of a maneuver.*
- *An audit logging module that stores commands and corresponding operations to be undone. It also stores commands in a Doubly Linked List with chronological order. When executed, the commands are stored in a permanent file that can only be appended to, collision_avoidance.log.*

Assumptions

Local Telemetry: We can assume that the spacecraft has access to its position and sensor data continuously even if it can't communicate with ground control.

Maneuver Physics: For preliminary assessments of how feasible evasive burns are, we will use the approximation of the constant mass ($t_{burn} = m\Delta V/F$).

Stable Hardware: We believe that the flight computer is capable of executing custom C++ queues without losing stability.

Model Shortcuts: We believe that our engineered ratio (d_{miss}/σ_{pos}) is sufficient to be used within Logistic Regression for modeling complex, non-linear aspects of orbital mechanics.

References

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